

Tidal Space: Interactive Home Installation for Work-From-Home Parents

Martina Huynh
Cream on Chrome, Rotterdam,
Netherlands
huynhmartina@gmail.com

Jonas Althaus
Cream on Chrome, Rotterdam,
Netherlands
jonas.althaus@protonmail.com

Hyunjung Kim
DLX Design Lab, The University of
Tokyo, Chiba, Japan
hyunjung.keem@gmail.com

ABSTRACT

The past few years of COVID-19 lockdown have made it abundantly clear that both childcare and professional work are inextricable from the home office – as many were forced to juggle between various roles and expectations while caring for their family and working from home. Tidal Space considers the manifold needs of work-from-home parents by incorporating motorized curtains and a foldable acoustic panel into the home office. Triggered by screen activity, the curtains self-adjust to serve as spatial moderators, mediating the boundaries between different types of work. From soundproof separation for focused work, to more translucent and open configurations for checking on the children, to interactive elements allowing kids and parents to play, Tidal Space aims to improve the home office experience for all.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design process and methods; Interface design prototyping.

KEYWORDS

New normal, Work from home, Home office, Kinetic installation, Programmable curtains

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1 INTRODUCTION

As working from home (WFH) has become the new normal, challenges have continued to face workers, especially those parenting young children. Recent studies on WFH during the COVID-19 pandemic have found that parents have been disproportionately negatively affected in their well-being and productivity [Machado et al., 2021; Ralph et al., 2020]. While many WFH parents appreciate being close to their children, they report challenges such as more distractions and interruptions, which leads to inefficient use of their work time [Ford et al., 2021].

Our work aims to redesign the home office space to solve the challenge of WFH parents being physically present with children but mentally focused on other tasks. Most home office solutions mainly focus on functional aspects; for example, setting up a place with more and better hardware (e.g., dual screens, powerful laptops) or ergonomic furniture (e.g., standing desk stools). However, working at home has an extra social aspect that must be considered, specifically, interacting with family members in their home office [Salazar, 2001]. Our goal is to build flexible boundaries between work, childcare, and home that suit the situational needs of WFH parents. Tidal Space's focus on building positive work-home interaction [Geurts et al., 2005] in post-pandemic homes sets it apart from other COVID-19-inspired projects recently exhibited at SIGGRAPH and SIGGRAPH Asia, which centered the technical aspects of promoting individual health [Pataranutaporn et al., 2020] and minimizing contact with each other [Akagawa et al., 2021; Hudnell and King, 2021; Iwasaki et al., 2021].

2 DESIGN CONCEPT

Tidal Space works to mediate a boundary that changes fluidly, offering different levels of spatial, visual, and acoustic separation between the home office and the home. Its design is based on the assumption that not all activities require a separation of work and family life. For example, writing papers requires full concentration and a quiet environment. However, tasks such as reading emails or downloading files allow some level of noise and disruption. We identified four main activities that Tidal Space should support:

- **Focused work:** work that demands intense concentration and requires a quiet environment (e.g., writing papers or proposals)
- **Communication work:** work that involves live audio/video communication and requires shielding from noise and visual clutter (e.g., online meetings/lectures/presentations, calls)
- **Light work:** work that can be done with relatively minimal attention; work that is open to minor distractions (e.g., browsing for inspiration, checking emails)
- **Childcare work:** daily caretaking activities that require a comfortable and playful environment (e.g., eating/playing together, chatting)

3 IMPLEMENTATION

Tidal Space consists of motorized Austrian curtains and a foldable acoustic panel suspended from the ceiling. The curtains are made of translucent chiffon textile that separates parents and children while allowing both to feel each other's presence. The arc-shaped acoustic panel is a fabric structure made of origami folded soundproofing felt, creating an environment for focused work when extended

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Figure 1: The extended acoustic panel behind the seat.

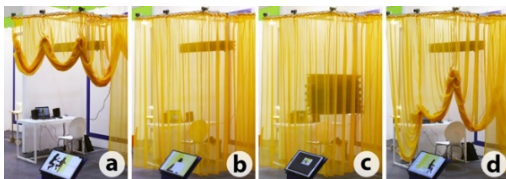


Figure 2: Tidal Space in (a) *Idle mode* (curtains up, acoustic panel folded), (b) *Lite mode* (curtains down, acoustic panel folded), (c) *Focused mode* (curtains down, acoustic panel extended), and (d) *Playtime mode* (curtains with programmed movement, acoustic panel folded).

(see Figure 1). Built-in stepper motors raise and lower the curtains and panel automatically. Figure 2 summarizes the actions of the curtains and panel in the four modes— a) *Idle*, b) *Lite*, c) *Focused*, and d) *Playtime*— to support different activities. In *Idle mode*, the curtains and panels are raised, so there is no separation. In *Lite mode*, the curtains fall to enable soft and translucent separation suitable for light work. In *Focused mode*, in addition to the settings of *Lite mode*, the acoustic panel is also extended. Lastly, in *Playtime mode*, the curtains continually move gently up and down, creating a welcoming and playful atmosphere for children and parents to play together during breaks from work. Since the majority of WFH workers use computers, Tidal Space plans to use screen activity as the main trigger for the different modes. For example, when the Zoom app is launched, Tidal Space turns *Focused mode* on; curtains and the soundproofing panels are automatically lowered to provide a visual and acoustic separation, as well as an appropriate background. Through the above features, Tidal Space aims to embody the vision of future smart homes presented by [Mennicken et al., 2014]: "homes to be context-aware domestic spaces" that implement automation to support individuals participating in domestic routines, while simultaneously "keeping people from becoming disengaged and allowing them to maintain important values."

4 DEMONSTRATION

Tidal Space will operate exactly like the three-and-a-half-minute video demonstrating how the installation supports WFH parents [Kim et al., 2021]. The video depicts a typical day of a WFH mother

with two children: a nine-year-old elementary school girl taking online classes at home and a five-year-old preschooler. It contains the mother's screen activities, online video meetings and email/digital whiteboard, which trigger *Focused* and *Light modes*, respectively. It also includes scenes of the parent having lunch with her daughter using *Idle mode* and the father playing with his children using *Playtime mode*. This video will be played on a loop on the monitor alongside the installation.

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